

CMS Open Data

teacher: Kati Lassila-Perini (will arrive Tuesday to work with you)

slides taken from Kati Lassila-Perini and Edgar Carrera Jarrin

BND Graduate School, Heimbach, 2026

MOTIVATION

■ Why open data: preservation done in public

1

Reproducible science

Publicly available data are a cornerstone of transparent, verifiable research.

2

Value beyond CMS

Theorists, external experimentalists, educators and citizen scientists can extract new science from the same data.

3

Preservation through use

Outside use keeps data, software and know-how alive, and exposes archival gaps while they can still be fixed.

First at the LHC

In November 2014, CMS became the first LHC experiment to release research-quality open data. The pipeline has run ever since.

THE MANDATE

Open access to data is a CERN mission value, reaffirmed in the 2020 European Strategy. CMS policy turns it into concrete, scheduled release rules (next slide).

POLICY & FRAMEWORK

■ What is released, and when

WHAT: FOUR DATA LEVELS

Level 1 · Publications

papers plus numerical data (HEPData, INSPIRE)

Level 2 · Outreach

simplified formats, 200+ derived datasets

Level 3 · Reconstructed

data + simulation + software + docs: the open-data
workhorse

Level 4 · Raw

custodial copies; small samples for ML

WHEN: THE LEVEL 3 RELEASE CLOCK

data taken

+6 years

+10 years



50% released

100% released, or
when analysis ends

Cap: at most 20% public while similar data
are still being taken.

*Levels 1 and 2 are released promptly, at
publication or as approved.*

DPHEP data-complexity levels, common to the LHC experiments

Released under CC0 with persistent DOIs. [CMS policy \(DOI\)](#) · [CERN Open Data Policy \(CERN-OPEN-2020-013\)](#) ·

[FAIR principles](#) (Findability, Accessibility, Interoperability, and Reuse of digital assets)

OPEN DATA CONTENT

■ Format evolution: each generation lowers the barrier

Data tier	Event size	CMSSW needed?
Reconstructed (RECO)	~3 MB	Yes
AOD (Run 1)	~500 kB	Yes
MiniAOD (Run 2)	~50 kB	Yes
NanoAOD (flat ROOT ntuple)	1-2 kB	No

4 PB+

collision data & simulation, all public on the CERN Open Data Portal

300+

collision datasets, plus 50 000+ simulated datasets

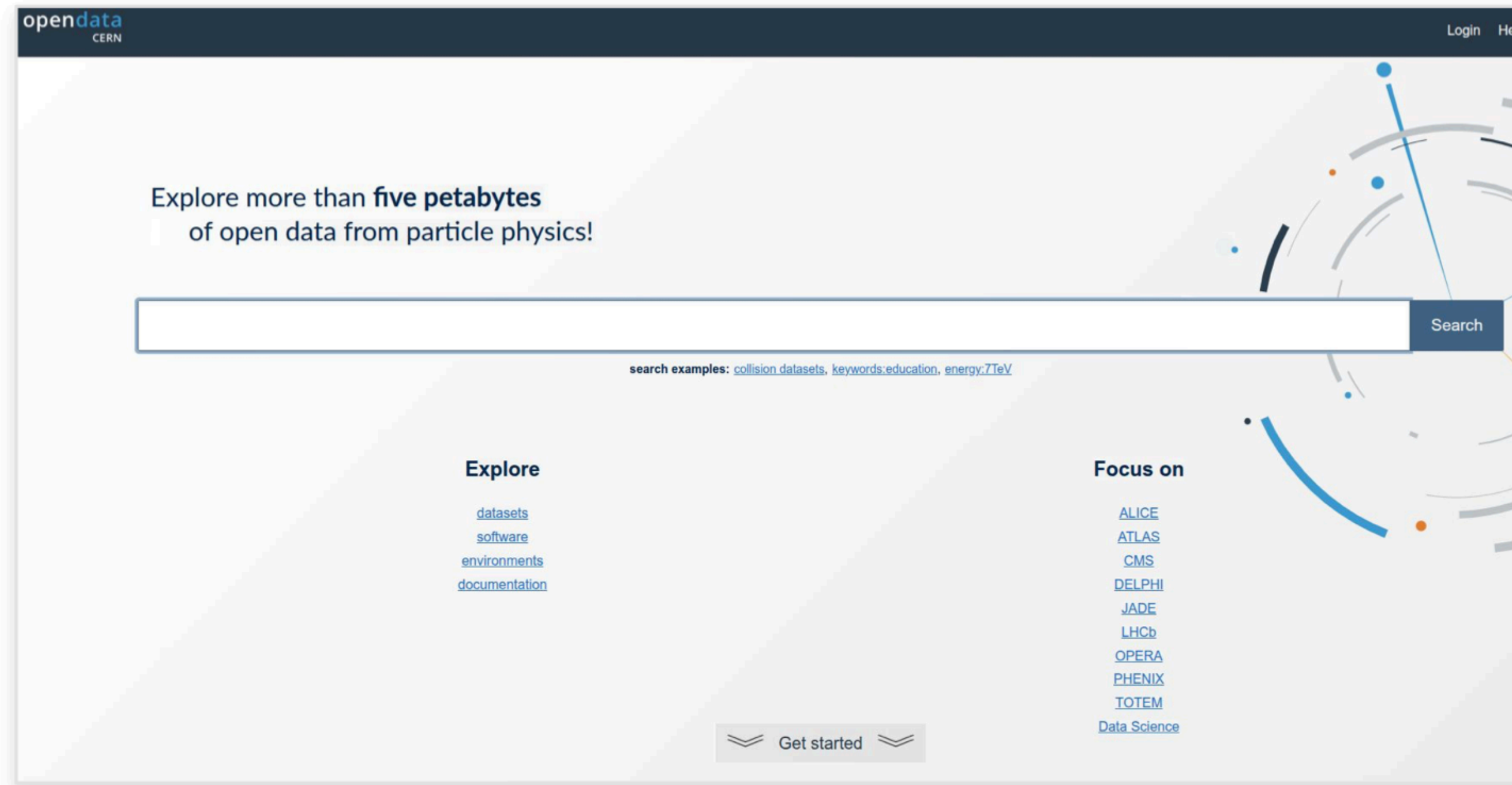
CMSSW: the CMS SoftWare framework (required to read the older tiers.)

NanoAOD (first open release April 2024, 2016 data): ~95% smaller than MiniAOD, plain ROOT files, no CMSSW. Analyze with Python: uproot, awkward, Coffea, RDataFrame.

PFNano: NanoAOD plus Particle Flow candidates, for jet studies and ML (e.g. Aspen Open Jets).

OPEN DATA ACCESS

■ Where the data live: the CERN Open Data Portal



SingleMuon primary dataset in NANOAOB format from RunH of 2016 (/SingleMuon/Run2016H-UL2016_MiniAODv2_NanoAODv9-v1/NANOAOB)

/SingleMuon/Run2016H-UL2016_MiniAODv2_NanoAODv9-v1/NANOAOB, CMS Collaboration

Cite as: CMS Collaboration (2024). SingleMuon primary dataset in NANOAOB format from RunH of 2016 (/SingleMuon/Run2016H-UL2016_MiniAODv2_NanoAODv9-v1/NANOAOB)
DOI:10.7483/OPENDATA.CMS.4BUS.64MV

There is one publication referring to these data

Data recorded in 2016. Published in 2024.

Dataset Collision CMS 13TeV pp CERN-LHC

Description

SingleMuon primary dataset in NANOAOB format from RunH of 2016. Run period from run number 281613 to 284044.

The list of validated runs, which must be applied to all analyses, either with the full validation or for an analysis requiring only muons, can be found in:

Every record carries a persistent, citable DOI.

FAIR (Findability, Accessibility, Interoperability, and Reuse of digital assets) IN PRACTICE

Datasets, software, environments and documentation, all findable in one place and citable by DOI.

The CERN Open Data Portal: categorised, searchable, browsable by experiment.

Explore it: opendata.cern.ch

■ Preserving environments, not just bytes

CMSSW in Docker containers & VMs

the CMS SoftWare framework, packaged to run on any OS



Conditions database

calibrations and alignments



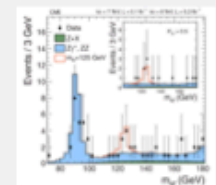
Validated run lists

certified good-quality data



Example workflows

e.g. the Higgs to 4-lepton discovery plot



Reproducible analysis

by anyone, on any machine, years after the data were taken

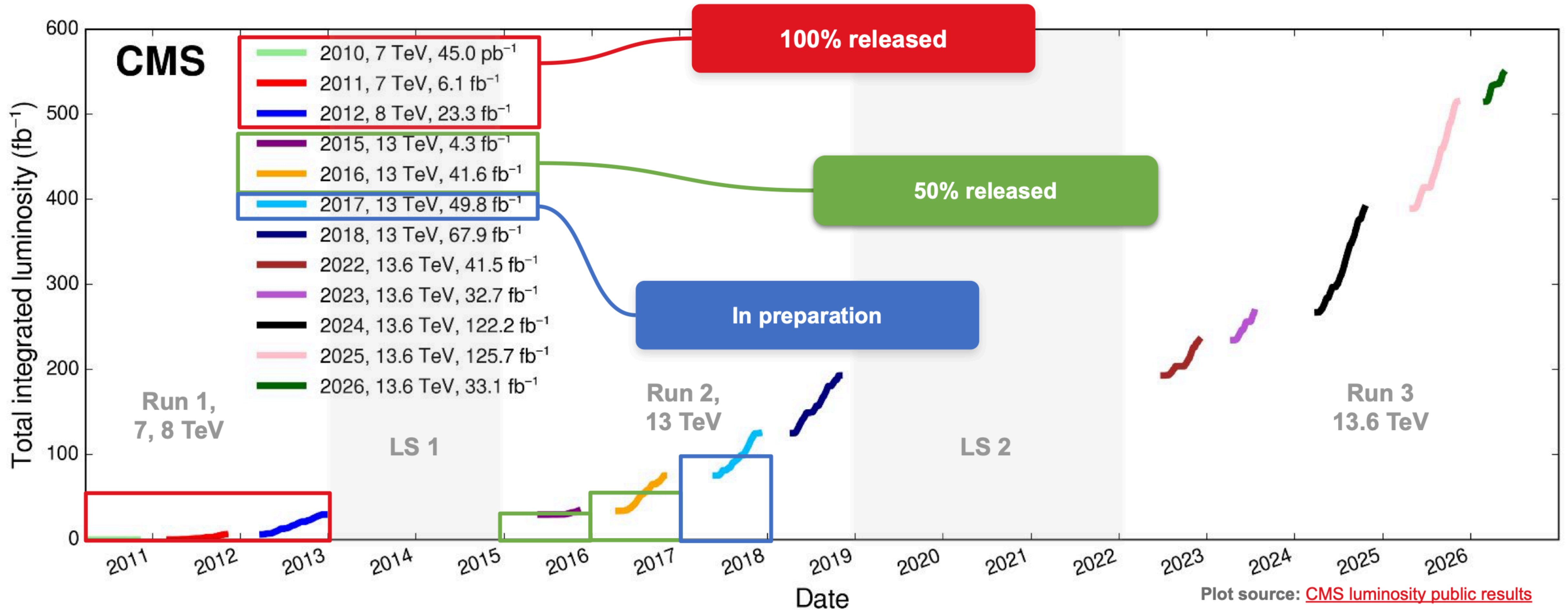
THE HARD PART

Keeping a full software stack alive and runnable by outsiders is the real preservation engineering.

Reality check: raw data (Level 4) are not practically usable outside the collaboration. Reconstruction is centralized even inside CMS.

Analysis documentation: cms-opendata-guide.web.cern.ch · Support: opendata-forum.cern.ch

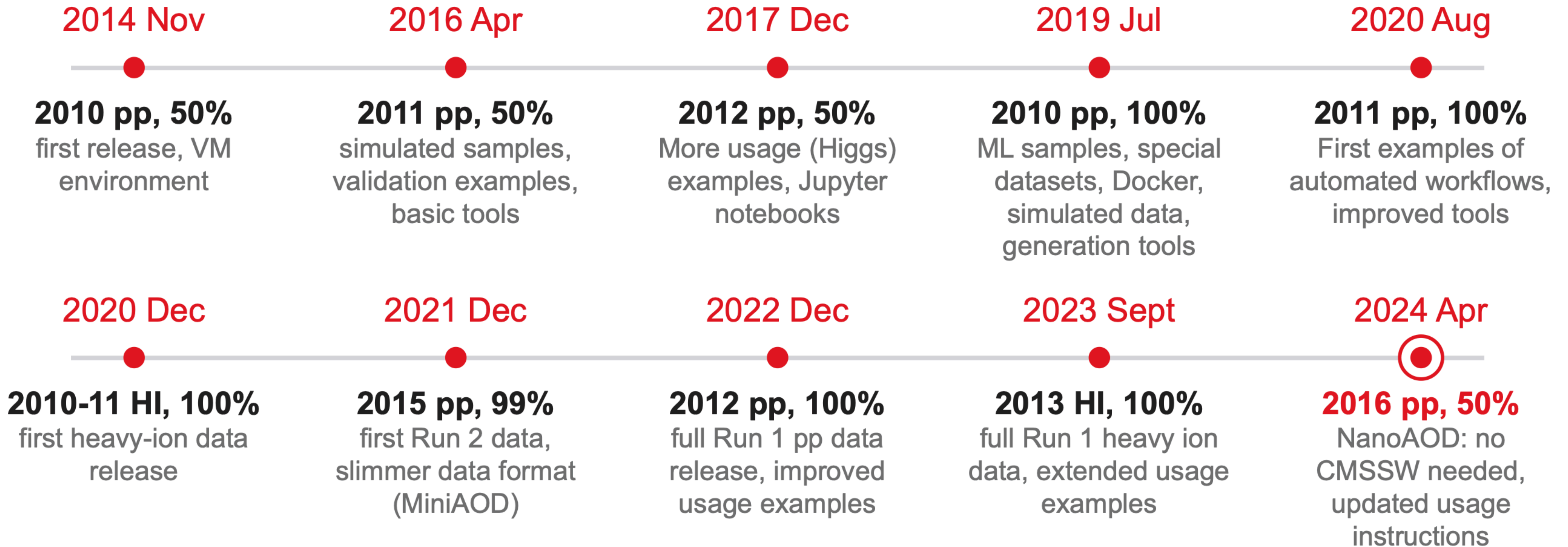
■ What is public, era by era



Cumulative pp luminosity delivered to CMS, with open-data release status overlaid per era.

SUCCESSSES · RELEASES

■ A sustained pipeline: ten releases in ten years



Next: 2017 pp. Release approved by the Collaboration Board; final arrangements are now being completed.

SUCCESSES · RESEARCH

Real physics by groups outside CMS

100+

papers citing CMS Open Data DOIs (INSPIRE, Jun 2026): mostly small groups, spanning physics results, new methods, and infrastructure R&D.

QCD · 2017

First novel analysis on open data: the QCD splitting function from jet substructure, 2010 AOD.

[arXiv:1704.05066](https://arxiv.org/abs/1704.05066)

SEARCH · 2021

Dark matter produced with a heavy Z' dimuon resonance, 2012 data.

[arXiv:2109.11274](https://arxiv.org/abs/2109.11274)

ML · 2025

Anomaly detection re-discovers Y mesons in non-isolated muon pairs, 2016 NanoAOD.

[arXiv:2502.14036](https://arxiv.org/abs/2502.14036)

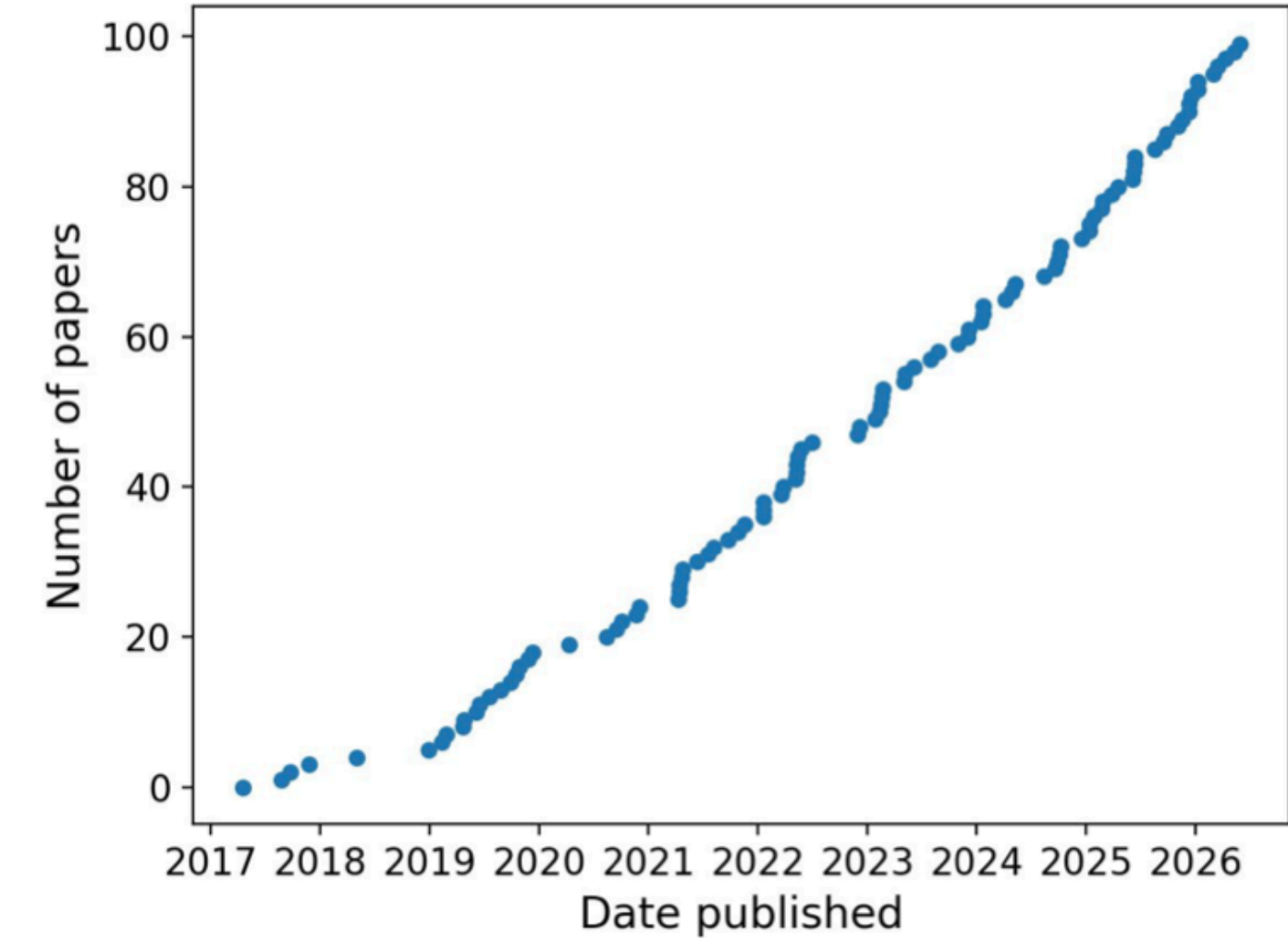
INFRASTRUCTURE

IRIS-HEP Analysis Grand Challenge: full ttbar pipeline on 2015 data, stress-testing HL-LHC analysis tools at scale.

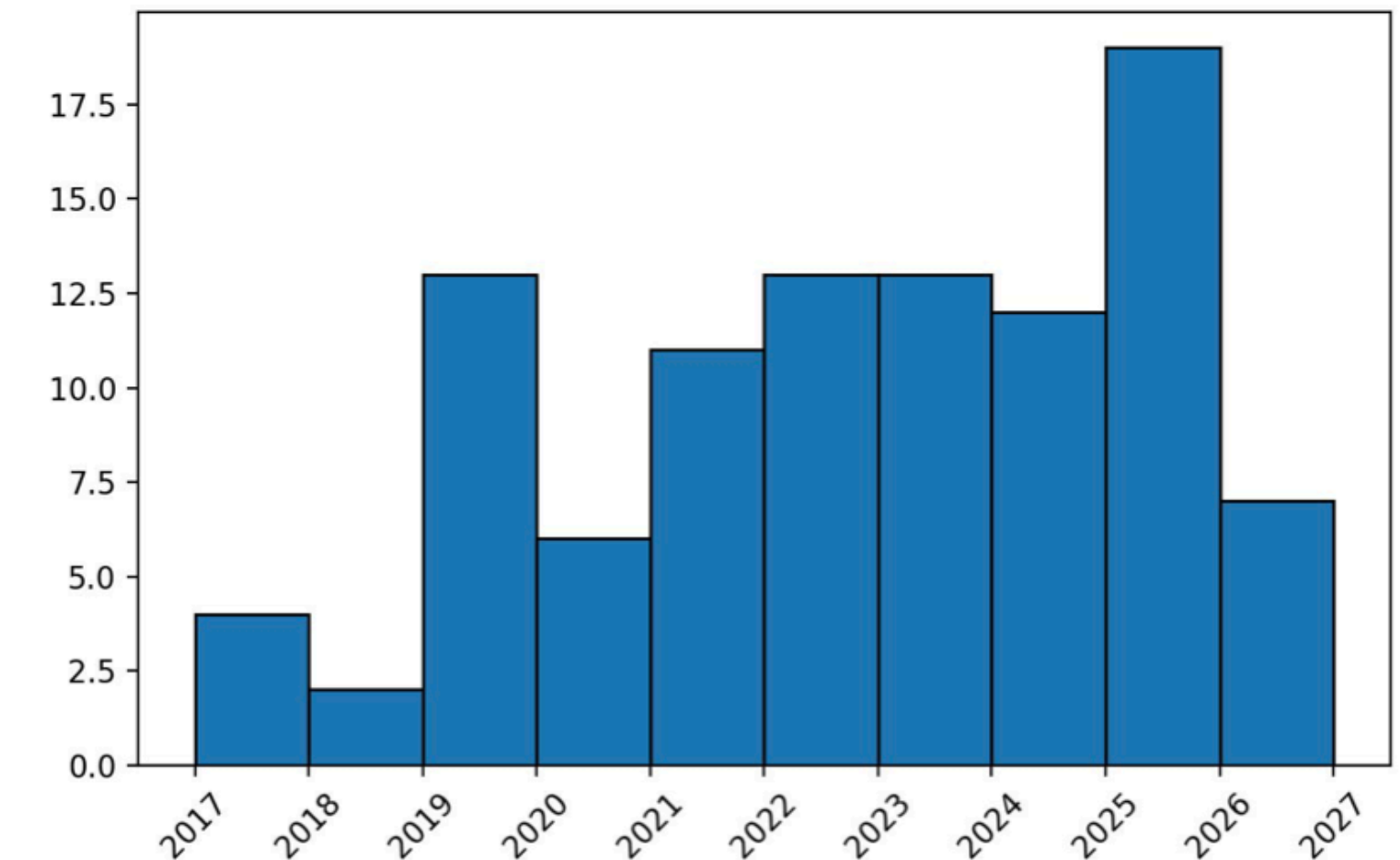
agc.readthedocs.io

Track them on INSPIRE: search references to DOIs **10.7483/OPENDATA.CMS***

Papers citing CMS Open Data DOIs [Inspire]
Excluding CMS/CERN open data teams and CMS Collaboration
2026-06-25



Papers citing CMS Open Data DOIs [Inspire]
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Publications citing CMS Open Data DOIs over time (CMS and CERN open-data authors excluded).

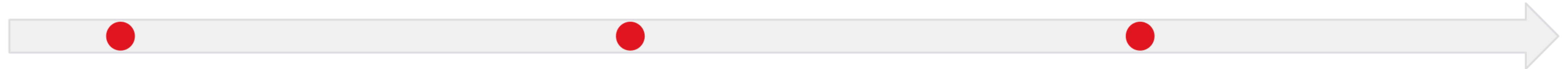
OPPORTUNITIES

■ The road ahead: 2026 to 2030

2026

2027-29

2030



Partial release of the 2017
dataset, in preparation
now

Remaining Run 2
releases; new NanoAOD
examples and guides

All of Run 2 public,
alongside the first Run 3
(13.6 TeV) releases

ML SUBSTRATE

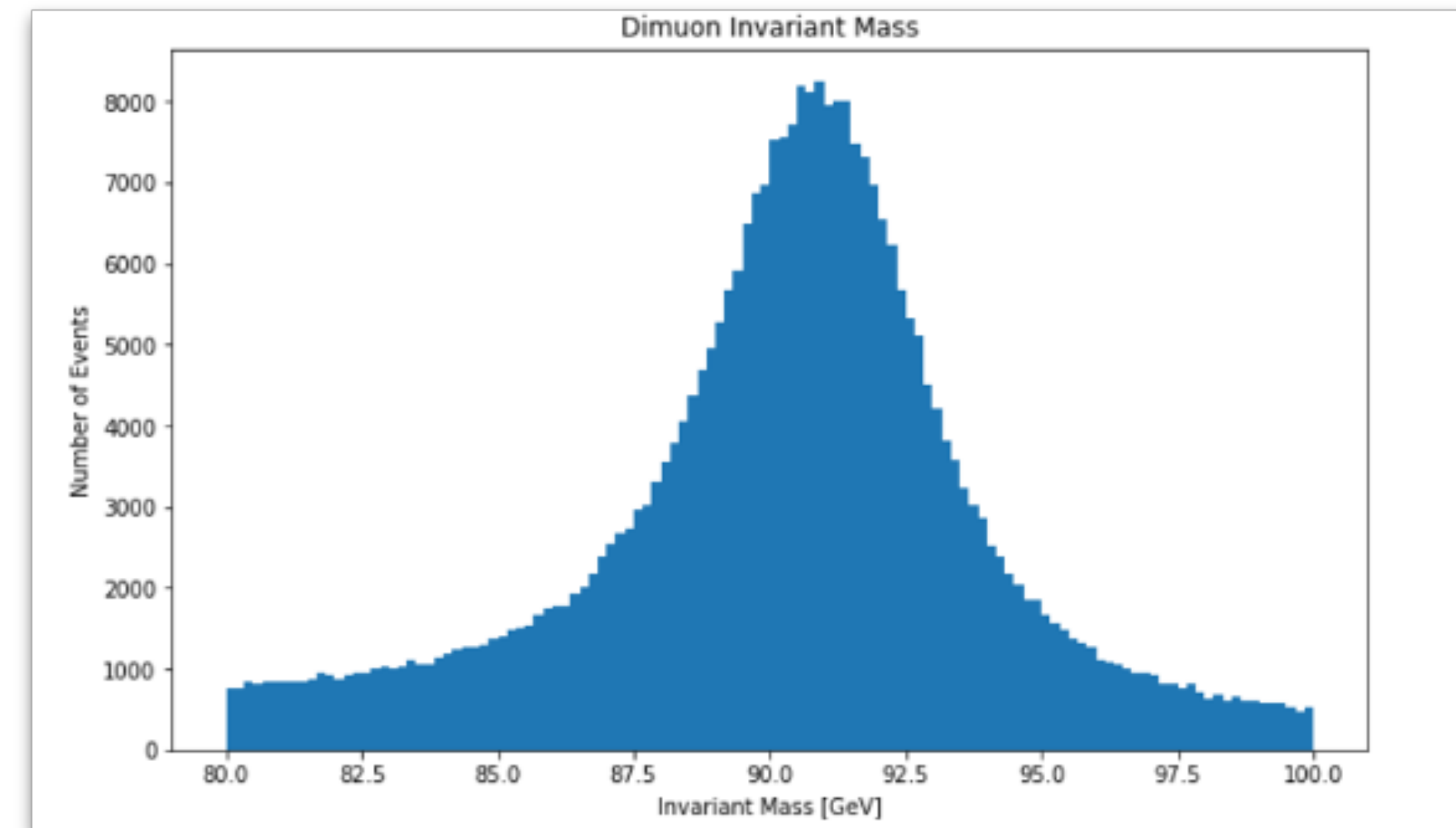
NanoAOD and PFNano meet the ML community where it lives: flat arrays, standard Python tools, streaming-friendly small files, no bulk downloads.

THE FRONTIER IS REUSE

The pipeline is mature. The leverage now is better tooling, lower barriers, and a larger community, not just more bytes.

First Task

- A quick start as an appetizer:
 - Reconstruct particles decaying into two muons from 2016 data (Z, Υ , J/Ψ ...)
- Think what kind of studies are possible (and interesting) using CMS Open data.
 - Available data? Available simulations?
- Explore how to get started.
 - Documentation? Tools?
- Estimate the time and resources needed for the study of your choice
 - Computing resources? Time?



Learning by Yourself

- CMS Open Data workshops are the best guide for getting started
 - [How to use 2016 CMS open data](#) (in the format not requiring CMS software)
 - The most adapt for research purposes
 - [CMS open data in education](#)
 - Some nice learning material for further understanding of particle physics data
- [Summary of resources and tools](#)
- Background knowledge
 - [CMS detector](#)
 - [Physics objects](#) in CMS data files
- Working environment and data access, on your laptop
 - install the needed [Python packages](#) preferably in a virtual environment, or install docker-engine (or Docker Desktop) and [use the available container](#)

Hands On

- [Find and explore datasets](#)
- [Di-muon reconstruction](#) (more in detail than the quick start)
 - no need to download the files, stream with xrootd
- [Learn about simulated data](#) ($W \rightarrow \mu\nu$)
- [template for a Python NanoAOD analysis with trigger, validated-run mask and simulation weights:](#)

Task to work on by yourself now:

- get yourself familiar with the linked material and tutorials
- then measure $\sigma(pp \rightarrow Z/\gamma^* \rightarrow \mu\mu)$ for $60 < m(\mu\mu) < 120$ GeV at 13 TeV with CMS 2016 open data, including syst uncertainty, and compare with the published CMS result.
- continue directly from the first task's dimuon peak, and reuse some of the linked material: dataset finding, XRootD streaming, the $W \rightarrow \mu\nu$ truth-level notebook
- bandwidth limitation: copy files locally, should fit on laptops